

Low-Thrust SEP and Aerobraking Trajectory Optimization for Earth-Mars Transfer

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Contents

1	Introduction	2
1.1	Motivation	3
1.2	Problem Statement	3
1.3	Objectives and Scope	4
1.4	Report Organization	4
2	SEP Background	5
2.1	Governing Equations and Models	5
2.2	Assumptions and Limitations	6
3	SEP Low-Thrust Optimization Approach	6
3.1	Problem Formulation	7
3.2	Numerical and Analytical Methods	7
3.2.1	Analytical Q-law	7
3.2.2	Hermite-Simpson Collocation Method	9
4	SEP Case Study	9
4.1	Spacecraft Configuration	10
4.2	Design Approach Details	10
4.2.1	Q-law Design Approach	11
4.2.2	Direct Collocation Design Approach	12
4.3	Simulation Setup	12
5	SEP Trajectory Optimization Results and Discussion	13
5.1	Impulsive Solution	14
5.2	Low-Thrust Solution	15
5.3	Sensitivity Analysis	17
5.3.1	Thrust Magnitude Variation	17
5.3.2	Specific Impulse Sensitivity	18
5.3.3	Launch Window Analysis	19
5.4	Interpretation	20

6	Aerobraking Trajectory Modeling	20
6.1	Introduction	20
6.2	Martian Orbit Path Selection	21
6.3	Modeling of Atmospheric Passes	21
6.4	Modeling J_2	23
6.5	Atmospheric Density Model	23
6.6	Dynamic Pressure and Thermal Constraints	24
7	Aerobraking Parametric Mass Study Results	25
7.1	Aerobraking Parametric Mass Study	25
7.2	Results and Discussion	26
8	Conclusions and Future Work	30
8.1	General Key Findings	30
8.2	SEP Future Work	31
8.3	Aerobraking Future Work	31
A	Heliocentric Modified Equinoctial Elements	34
B	Q-law parameters	34

Nomenclature

m	=	Spacecraft mass [kg]
t	=	Time [s]
T	=	Total maneuver time [s]
ΔV	=	Delta V [m/s]
v	=	Velocity [m/s]
r	=	Radial distance from planet center [m]
ρ	=	Atmospheric density [kg/m ³]
C_D	=	Drag coefficient
A	=	Reference area [m ²]
D	=	Drag force [N]
μ	=	Mars gravitational parameter [m^3/s^2]
g	=	Gravitational acceleration [m/s ²]
a	=	Acceleration [m/s ²]

1 Introduction

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1.1 Motivation

Traditional interplanetary missions have primarily relied on chemical propulsion, which provides high thrust but is limited by low fuel efficiency. As mission requirements evolve to include heavier payloads and more distant targets, the mass penalty associated with carrying large quantities of chemical propellant becomes a more significant constraint. Solar Electric Propulsion (SEP) offers a solution by utilizing electricity from solar arrays to accelerate propellant to high exhaust velocities, achieving a specific impulse (Isp) nearly ten times that of chemical rockets.

Concurrently, aerobraking has proven to be an effective strategy for reducing propulsive requirements during planetary orbit insertion, leveraging atmospheric drag. Mars Global Surveyor (1997) was the first successful Martian mission to use aerobraking, using its large solar panels as drag surfaces over months of atmospheric passes around Mars. Though it took around 130 Earth days, it reduced propellant requirements by $\sim 75\%$, significantly increasing payload capacity for the mission. In 2014, the MAVEN mission also successfully used aerobraking to transition from an elliptical orbit into a lower science orbit.

Both these techniques have largely been applied independently and primarily to smaller, satellite-scale missions. As Mars missions continue to increase in scale and complexity, propulsion efficiency remains a key limitation in mission design. Larger payloads and more demanding orbital requirements increase the amount of propellant required, reducing the mass available for payload. In response, this project explores a combined mission architecture that uses Solar Electric Propulsion (SEP) for a Earth-Mars transfer and aerobraking during Mars orbit insertion. Our overall goal is to evaluate whether pairing these two methods can significantly reduce total propulsive requirements compared to a traditional chemical transfer and insertion approach, and calculate feasible trajectories.

1.2 Problem Statement

The first part of this project is focused on optimizing a low thrust, SEP, trajectory from Earth to Mars, using NASA’s Psyche spacecraft as a base model to analyze the transition from impulsive to low-thrust trajectories. It is best split into two objectives: (1) to design and optimize a low-thrust Earth-Mars transfer, comparing the total Δv and mass requirements to a standard Hohmann transfer, and (2) to calculate and present a Mars orbital insertion trajectory for a future Martian spacecraft that utilizes aerobraking to decrease Δv requirements. By simulating continuous-thrust maneuvers within a heliocentric framework, this study quantifies the efficiency gains and mission duration trade-offs inherent in SEP systems. We will produce a set of optimized trajectories that demonstrate how SEP reduces the total propellant mass fraction.

The second part of this paper begins at the end of the SEP transfer and focuses on optimizing a Mars orbit insertion trajectory using aerobraking. A Python-based numerical simulation is used to propagate repeated atmospheric passes, allowing drag to reduce orbital energy in place of a fully propulsive capture maneuver. This analysis outputs Δv and fuel savings, considering key constraints such as thermal loading, dynamic pressure, atmospheric density, and required correction burns, with the ultimate goal of assessing the efficiency of using aerobraking to enter the Martian atmosphere.

1.3 Objectives and Scope

The first primary objective of this research is to quantify the performance advantages of SEP over traditional chemical propulsion for an Earth-Mars transfer. By using the Psyche spacecraft’s specifications as a baseline, this study evaluates the reduction in propellant mass and the specific Δv requirements for a low-thrust trajectory. Additionally, the project assesses the sensitivity of the optimized path to variables such as available thrust levels, specific impulse, and planetary launch windows to determine how these factors influence mission feasibility.

The scope of this project involves the implementation of several key modeling and optimization components. Initial work focuses on establishing a baseline through the calculation of an impulsive Hohmann transfer, including the necessary plane change to match Martian inclination. For the low-thrust analysis, we aimed to use a direct method to find a highly optimal solution, which required us to provide an initial seed trajectory. As a result, we chose to implement a Q-law controller for closed-loop initial trajectory generation and Hermite-Simpson collocation for direct transcription optimization. Hermite-Simpson was chosen as a medium-order method since higher-order optimization techniques were not necessary for the scope of this project. Performance is measured using four primary metrics: total Δv , final propellant mass fraction, time of flight, and the convergence efficiency of the optimization algorithms.

The second primary objective of this paper is to evaluate aerobraking trajectories following the SEP transfer. This section of the paper has three main objectives: (1) to calculate and optimize a nominal aerobraking trajectory for a spacecraft entering Martian orbit, (2) conduct a parametric study exploring how effective aerobraking is at Mars for different spacecraft configurations, and (3) calculate metrics for trajectory selection (fuel consumption, transfer time, number of passes) to compare the effectiveness of an aerobraking trajectory to a fully propulsive insertion trajectory.

This analysis includes a numerical propagation model in Python that accounts for Martian gravity, atmospheric drag, and J_2 of Mars. It also considers and analogizes the effects of atmospheric density, spacecraft mass, drag area, peak dynamic pressure constraints, heat flux constraints, and drag coefficient on aerobraking performance. The scope of this project however is limited to a reduced-order, "pass by pass" simulation rather than a full numerical propagation. The analysis also is designed to approximate the density profile of Mars, simplified spacecraft geometry, and ideal periapsis targeting on each pass. As a result, the model is meant to provide insight into comparing aerobraking performance across spacecraft configurations, rather than providing a final trajectory.

1.4 Report Organization

This report is organized into five major sections designed to provide a comprehensive analysis of interplanetary transfer efficiency. **Section 2** establishes the theoretical foundation for the study, detailing the mechanics of continuous and non-continuous low-thrust SEP in contrast to traditional impulsive maneuvers. This section also introduces the conceptual framework of the chosen trajectory optimization methods and the governing principles of the baseline Hohmann transfer, including necessary inclination changes.

Section 3 describes the mathematical basis of the trajectory optimization techniques employed, specifically focusing on the implementation of Lyapunov-based control and Hermite-Simpson collocation. It further outlines the computational tools, such as NASA’s EMTG, utilized to execute the numerical simulations. **Section 4** and **Section 5** detail the practical application of this approach to a simulated transfer from Earth’s Low Earth Orbit (LEO) to Mars using the Psyche spacecraft as a primary design reference. These sections provide a comparative evaluation of impulsive versus low-thrust strategies and a sensitivity analysis of various control parameters, such as thrust-to-power ratios and launch epochs.

The final segments of the paper transition to the Martian orbit insertion phase. **Section 6** introduces aerobraking trajectory modeling, focusing on how a spacecraft can use atmospheric drag to achieve a final circularized orbit rather than a purely propulsive burn. **Section 7** presents the results of a parametric mass study that identifies which spacecraft characteristics, such as mass and drag area, make aerobraking more efficient. Finally, **Section 8** concludes the report by synthesizing key findings on propellant savings and suggesting methods to enhance the robustness of the solution, such as the integration of n-body dynamics.

2 SEP Background

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The fundamental advantage of SEP over chemical propulsion lies in its high specific impulse, which significantly reduces the propellant mass required for deep-space missions. While chemical engines provide high thrust through brief, high-energy combustion, SEP systems utilize electrical energy to accelerate propellant to much higher exhaust velocities. This enables missions that are otherwise impractical or impossible to conduct using traditional methods. The Psyche spacecraft serves as a contemporary model for this technology, utilizing a high-power SEP architecture derived from commercial geostationary (GEO) satellite heritage. Specifically, the mission employs the SPT-140 Hall thruster, a system qualified for commercial use that has been modified to operate at solar distances of up to 3.3 AU [1].

Optimizing these low-thrust trajectories requires a shift from classical orbital mechanics to optimal control theory. In a traditional impulsive Hohmann transfer, maneuvers are modeled as instantaneous changes in velocity Δv at specific points in a conic section orbit. SEP maneuvers, conversely, involve a continuous steering problem where the thrust vector must be determined at every point along the path to minimize fuel consumption or time of flight. This project utilizes Q-law control to generate initial trajectory guesses, which will be refined with collocation.

2.1 Governing Equations and Models

The spacecraft’s motion is modeled within the framework of the heliocentric two-body problem, where the Sun is the primary gravitational source. This approach simplifies the control laws for implementation while maintaining sufficient accuracy for conceptual mission design. For the baseline impulsive case, the Hohmann transfer equations provide the Δv benchmark.

For the SEP model, the standard equations of motion are modified to include a continuous thrust acceleration term

$$\frac{d\mathbf{r}}{dt} = \mathbf{v} \quad (1)$$

$$\frac{d\mathbf{v}}{dt} = -\frac{\mu}{r^3}\mathbf{r} + \frac{\mathbf{T}}{m} \quad (2)$$

The system state includes the instantaneous mass m , which decreases according to the flow rate

$$\frac{dm}{dt} = -\frac{\|\mathbf{T}\|}{c} \quad (3)$$

where c is the effective exhaust velocity $g_0 I_{sp}$. The power-to-thrust relationship is a critical factor in which available power scales with the inverse square of the distance from the Sun ($1/r^2$).

2.2 Assumptions and Limitations

The trajectory analysis and numerical simulations in this study are defined by two key assumptions designed to maintain computational feasibility:

1. The primary simplification is the use of heliocentric two-body dynamics, which models the spacecraft's motion exclusively under the gravitational influence of the Sun. While an n-body approach provides better accuracy by accounting for the gravitational effects of other major bodies and solar radiation pressure, the two-body model is sufficient for conceptual engineering and reduces the complexity of the control laws.
2. For the baseline Hohmann transfer, the orbits of Earth and Mars are treated as circular to allow for a direct analytical comparison of impulsive Δv requirements and necessary inclination changes.

The project is also constrained by several limitations regarding the precision and scope of the results. The convergence of the final trajectory is highly sensitive to the initial guess; while the Q-law provides a feasible starting point, the resulting solution may represent a local rather than a global optimum.

This study also excludes planetary gravity assists to better isolate the performance comparison between SEP and chemical propulsion, even though the actual Psyche mission utilizes a Mars flyby to enhance mass margins. Furthermore, the analysis does not account for detailed spacecraft characteristics such as reaction control system (RCS) mass flow, attitude control transients, or power subsystem degradation. Finally, the study utilizes fixed thrusting and coasting arcs rather than a fully variable-throttle engine model, focusing on the broader efficiency trade-offs between the two propulsion strategies.

3 SEP Low-Thrust Optimization Approach

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3.1 Problem Formulation

The interplanetary transfer problem is set up as a low-thrust optimal control problem. The spacecraft motion is governed primarily by two-body gravitational forces and mass variation from propellant expenditure. The state vector is defined as

$$\mathbf{y} = [\mathbf{r}, \mathbf{v}, m]^T \quad (4)$$

where $\mathbf{r}$ and $\mathbf{v}$ are position and velocity vectors in the heliocentric inertial frame and m is the instantaneous spacecraft mass.

The spacecraft equations of motion are expressed as

$$\frac{d\mathbf{r}}{dt} = \mathbf{v} \quad (5)$$

$$\frac{d\mathbf{v}}{dt} = -\frac{\mu}{r^3}\mathbf{r} + \frac{T}{m}\mathbf{u} \quad (6)$$

$$\frac{dm}{dt} = -\frac{T}{I_{sp}g_0} \quad (7)$$

where μ is the gravitational parameter of the central body, I_{sp} is the specific impulse of the spacecraft thrusters, and g_0 is the standard gravitational acceleration. The control variables are T , the magnitude of thrust, and $\mathbf{u}$, the direction of thrust.

For optimal control, the design objectives are:

1. Minimize propellant consumption Δv
2. Minimize transfer time t_f

Consequently, the optimization problem operates under a cost function that minimizes

$$J = w_1\Delta v + w_2t_f \quad (8)$$

where w_1 and w_2 are tuned weights that balance the two objectives.

Boundary conditions for this problem are specified as departure from a 400 km altitude circular Earth parking orbit and arrival at a 300 km altitude circular Mars parking orbit. Initial mass and thrust capabilities correspond to those of the Psyche mission propulsion system.

3.2 Numerical and Analytical Methods

3.2.1 Analytical Q-law

Classical Q-law control is implemented in Python to rapidly generate initial guess trajectories. It minimizes a penalty function Q that represents error between the spacecraft's osculating orbit and a set of desired orbital elements. The result of this control law is thrust and coast segments that reduce this error over time. The initial trajectory design results from Q-law guidance that targets the osculating orbit of Mars at a specified arrival epoch.

The Q function is given by:

$$Q = (1 + W_P P) \sum_x S_x \left(\frac{x - x_T}{D_x} \right)^2, \quad \mathbf{x} = p, f, g, h, k, L \quad (9)$$

where x and x_T are the current and target orbital elements, respectively [2, 3]. The terms D_x represent the maximum rates of change for each orbital element, assuming optimal thrust, anywhere in the current osculating orbit. Elements with smaller values are "harder to change" and given greater weight in the Q function [4].

$$w = 1 + f \cos L + g \sin L \quad (10)$$

$$D_x = \begin{bmatrix} \max_L \left[\frac{2p}{\sqrt{\mu p}} \sqrt{(f \sin L - g \cos L)^2 + w^2} \right] \\ \max_L \left[\sqrt{\frac{p}{\mu}} \sqrt{\sin^2 L + ((w + 1) \cos L + f)^2} \right] \\ \max_L \left[\sqrt{\frac{p}{\mu}} \sqrt{\cos^2 L + ((w + 1) \sin L + g)^2} \right] \\ \max_L \left[\sqrt{\frac{p}{\mu}} \frac{s^2}{2w} |\cos L| \right] \\ \max_L \left[\sqrt{\frac{p}{\mu}} \frac{s^2}{2w} |\sin L| \right] \end{bmatrix} \quad (11)$$

The terms S_x are shaping functions for the prioritization of different elements. S_p ensures early correction of p by highlighting large errors and converges back to 1 near the target to discourage oscillations.

$$S_p = \left(1 + \left(\frac{|p - p_T|}{3p_T} \right)^4 \right)^{1/2}, \quad S_{f,g,h,k} = 1 \quad (12)$$

Additionally, this approach employs a penalty term

$$P = \exp \left(1 - \frac{r_p}{r_{p,\min}} \right) \quad (13)$$

which restricts trajectories with low altitude at perigee via the periapsis radius r_p and the minimum allowable periapsis radius $r_{p,\min}$, which is set to be the target periapsis radius in this simulation [2, 3].

The weights w_x are tuned for each orbital element and emphasize thrusting near periapsis for improved efficiency.

The relation between orbital element evolution $\dot{x}$ and optimal thrust direction $\mathbf{u}$ is given by Gauss' variational equations

$$\dot{x} = B\mathbf{u} \quad (14)$$

where B is a matrix describing the sensitivity of each orbital element to the perturbing thrust components. This matrix represents how Q changes with small variations in each element

and directs the thruster toward the most efficient direction [4].

$$\mathbf{B} = \begin{bmatrix} 0 & \frac{2p}{\sqrt{\mu p}} & 0 \\ \sqrt{\frac{p}{\mu}} \sin L & \sqrt{\frac{p}{\mu}} \frac{(w+1) \cos L + f}{w} & -\sqrt{\frac{p}{\mu}} \frac{g(h \sin L - k \cos L)}{w} \\ -\sqrt{\frac{p}{\mu}} \cos L & \sqrt{\frac{p}{\mu}} \frac{(w+1) \sin L + g}{w} & \sqrt{\frac{p}{\mu}} \frac{f(h \sin L - k \cos L)}{w} \\ 0 & 0 & \sqrt{\frac{p}{\mu}} \frac{s^2}{2w} \cos L \\ 0 & 0 & \sqrt{\frac{p}{\mu}} \frac{s^2}{2w} \sin L \end{bmatrix} \quad (15)$$

For optimal thrust components in the RTN frame $\mathbf{u} = [u_r, u_t, u_n]^T$, the control law is given by

$$\mathbf{D} = \begin{bmatrix} D_1 \\ D_2 \\ D_3 \end{bmatrix} = \mathbf{B}^T \nabla Q, \quad \mathbf{u} = \begin{bmatrix} u_r \\ u_t \\ u_n \end{bmatrix} = -\frac{\mathbf{D}}{\|\mathbf{D}\|} \quad (16)$$

3.2.2 Hermite-Simpson Collocation Method

For trajectory refinement and direct optimization, we use Hermite-Simpson collocation. The continuous-time optimization problem is transcribed into a discrete nonlinear programming (NLP) problem by dividing the trajectory into N segments with collocation points at nodes.

The Hermite-Simpson method enforces dynamic feasibility at segment midpoints via cubic Hermite interpolation. For each segment i , the state at the midpoint $t_{i+1/2}$ is approximated as:

$$\mathbf{x}_{i+1/2} = \frac{1}{2}(\mathbf{x}_i + \mathbf{x}_{i+1}) + \frac{h_i}{8}(\dot{\mathbf{x}}_i - \dot{\mathbf{x}}_{i+1}) \quad (17)$$

where $h_i = t_{i+1} - t_i$ is the segment time step, and $\dot{\mathbf{x}}_i = \mathbf{f}(\mathbf{x}_i, \mathbf{u}_i, t_i)$ represents the dynamics evaluated at node i .

The collocation constraint enforces that the dynamics at the midpoint must be consistent with the state derivative obtained from Simpson's quadrature:

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \frac{h_i}{6}(\dot{\mathbf{x}}_i + 4\dot{\mathbf{x}}_{i+1/2} + \dot{\mathbf{x}}_{i+1}) \quad (18)$$

where $\dot{\mathbf{x}}_{i+1/2} = \mathbf{f}(\mathbf{x}_{i+1/2}, \mathbf{u}_{i+1/2}, t_{i+1/2})$. This constraint is imposed at every segment, resulting in $7N$ equality constraints for the 7 -element state vector $[\mathbf{r}, \mathbf{v}, m]$.

The NLP decision variables consist of state vectors at all $N + 1$ nodes ($\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N$), control vectors at all nodes and midpoints ($\mathbf{u}_i$ and $\mathbf{u}_{i+1/2}$), and the time-of-flight t_f .

Boundary conditions constrain the initial state to match Earth's orbit and the final state to achieve Mars insertion. Additional path constraints enforce thrust magnitude limits $|\mathbf{T}| \leq T_{\max}$ and non-negative mass $m \geq m_{\text{dry}}$.

4 SEP Case Study

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4.1 Spacecraft Configuration

The spacecraft is configured with the main characteristics of Psyche, focusing on thruster capabilities of the SEP system.

Similar to Psyche’s SEP system, the simulated spacecraft Hall thrusters’ thrust depend on power level rather than a fixed thrust limit. Power availability scales roughly with inverse-square distance from the Sun, so power generated by the solar arrays on Earth P_{1AU} is used to determine input power throughout the transfer.

$$P_{in} = P_{1AU} \left(\frac{1AU}{r} \right)^2 \quad (19)$$

Thrust rises with input power but is clipped at the hardware limits of the SPT-140 thrusters.

$$T_{max} = \frac{2P_{in}}{v_e} \quad (20)$$

where c is the exhaust velocity, a characteristic of the propellant. The table below shows the Psyche-like parameters used for our simulation spacecraft [6, 1]. The notable change is to total mass, which has been increased due to higher necessary fuel budget for the Q-law low-thrust transfer compared to Psyche’s actual fuel budget.

Table 1: Spacecraft Parameters

Parameter	Value
Dry mass	1648 kg
Total mass	5000 kg
Max thrust	0.240 N *4
Specific impulse	1800 s
Solar power at 1 AU	20 kW
Thruster power range	0.9-4.5 kW

The starting Earth orbit is a circular 400km orbit, while the ending Mars orbit is a circular 300km orbit. The transfer window will be determined via Earth-Mars synodic geometry constraints. Launch opportunities in the late 2022 to early 2023 window preceding Psyche’s launch in October 2023 are tested.

4.2 Design Approach Details

In order to find a globally optimal solution in the chosen 2022-2023 search window, we first use Q-law control to analytically solve for an interplanetary transfer trajectory. Q-law control is an indirect method and cannot guarantee global optimality, but provides closed-loop, feedback-based guidance that chooses the most optimal control at every integration step. Treating this locally optimal trajectory as an initial seed, we can input the Q-law ”guess” into direct collocation, which discretizes the trajectory and further optimizes it for fuel efficiency and transfer time [7].

4.2.1 Q-law Design Approach

We chose to implement a Q-law controller using Modified Equinoctial Elements (MEE) primarily to avoid singularities that arise from equatorial (i=0) or circular (e=0) orbits, allowing for better numerical convergence. Since both departure and arrival orbits are circular, MEE is suitable. See Appendix A for definitions of orbital elements in relation to HCI.

The throttle law was formed based on the rate of change of the Q function. The goal of Q-law is to decrease Q over time; if $\dot{Q}$ is greater than a predetermined coast threshold, the controller outputs full throttle (1.0) because Q is worsening. If $\dot{Q}$ is less than the coast threshold negated, the controller outputs zero throttle and the spacecraft coasts. Otherwise, the controller outputs a nominal throttle command to steadily improve the shape match. This throttle value is fed into the thrust-power computation to determine how aggressively the available solar power should be used by the thrusters.

The controller operates in two stages: shape matching and phasing. During shape matching, the controller targets Mars' orbital elements, excluding true longitude L , which specifies a position along the orbit. When the spacecraft's osculating orbital elements reach within 3 percent of the target, the controller switches to phasing. In order to match L , a biased semi-major axis is calculated and input to the target at each timestep. The wrapped longitudinal error is given by

$$\Delta L = \text{mod}(L - L_T + \pi, 2\pi) - \pi \quad (21)$$

The target mean motion n_T gives rise to the time-to-go t_r , which describes the time needed to close the gap between the spacecraft's mean motion and the target.

$$n_T = \sqrt{\frac{\mu}{a_T^3}} \quad (22)$$

$$\Delta n_{\text{ref}} = \frac{3}{2} \frac{n_T}{a_T} 0.03 a_T \quad (23)$$

$$t_r = \text{clip}\left(\frac{|\Delta L|}{\Delta n_{\text{ref}}}, t_{\min}, t_{\max}\right) \quad (24)$$

From these equations, the instantaneous desired drift rate n_{des} is calculated and gives rise to the biased semi-major axis a_T^{biased} .

$$n_{\text{des}} = n_T - \frac{\Delta L}{t_r} \quad (25)$$

$$a_T^{\text{biased}} = \left(\frac{\mu}{n_{\text{des}}^2}\right)^{1/3} \quad (26)$$

Once the spacecraft reaches within a specified tolerance of the true target, thrusters are turned off and the simulation relies on natural orbital motion to further reduce orbital error. The Q-law trajectory is considered converged at the arrival epoch if the spacecraft state is within the arrival conditions, 0.5% of the true (unbiased) target and 5 degrees of longitude. A brief summary of chosen Q-law parameters is given in Appendix B.

4.2.2 Direct Collocation Design Approach

The direct collocation approach formulates the Earth-Mars transfer as a constrained non-linear optimization problem solved via Hermite-Simpson transcription (Section 3.2.2). The objective is to minimize propellant consumption, equivalently formulated as maximizing the final spacecraft mass m_f . The cost function is expressed as $J = -m(t_f)$, subject to the two-body equations of motion with continuous thrust and the collocation constraints that enforce dynamic feasibility.

Initial guesses for the collocation solver are generated using Q-law Lyapunov feedback control. The Q-law controller propagates a feasible trajectory from Earth orbit to Mars orbit by greedily descending the proximity quotient Q in modified equinoctial element space. This Q-law trajectory is then converted to Cartesian coordinates and resampled at the collocation nodes to provide an initial guess for the state and control histories. While Q-law does not guarantee optimality, it produces physically reasonable trajectories that satisfy the boundary conditions and lie within the feasible region, thus improving the convergence of the NLP solver.

Boundary conditions constrain the initial state $\mathbf{x}_0$ to match Earth’s departure orbit (circular at 1 AU) and the final state $\mathbf{x}_N$ to achieve Mars orbit insertion. Path constraints enforce the thrust magnitude limit $|\mathbf{T}| \leq T_{\max}$ at all nodes, where $T_{\max} = 5.0$ kW corresponds to the Psyche spacecraft’s Hall-effect thruster capability. Additionally, the spacecraft mass is constrained to remain above the dry mass throughout the trajectory: $m(t) \geq m_{\text{dry}}$ for all $t \in [0, t_f]$.

The collocation problem is solved using interior-point optimization methods, with 101 nodes selected to balance accuracy and computational cost. Convergence is declared when the constraint violation falls below 10^{-6} and the optimality conditions are satisfied to a relative tolerance of 10^{-5} .

This hybrid strategy combines the strengths of both methods: Q-law provides rapid feasible solutions and global exploration, while collocation ensures local optimality and precision. The approach balances computational efficiency with optimality for the Earth-Mars transfer problem.

4.3 Simulation Setup

Propagation of motion in this simulation is accomplished in two ways. First, the spacecraft is tracked in heliocentric inertial (HCI) coordinates. Second, heliocentric Earth and Mars states are loaded from offline ephemeris data from the JPL Horizons System [8].

Since it uses two-body equations of motion, this simulation also incorporates spheres of influence (SOI) for each central body. A switching mechanism is applied within the optimization such that the spacecraft is influenced only by the dominant gravitational body. For example, within Earth’s SOI, the spacecraft is subject only to Earth’s gravity. Once it exits Earth’s SOI, the spacecraft is influenced solely by the Sun until it enters Mars’ SOI.

Several parameters are required to configure the direct optimization and subsequent sensitivity analysis. One major design choice is the number of collocation nodes, which may be limited by computational cost. In this project, we use 101 nodes. Additionally, spacecraft thrust capabilities can be varied across different configurations.

To determine a search window of departure epochs near Psyche’s launch, porkchop plots of Δv were generated.

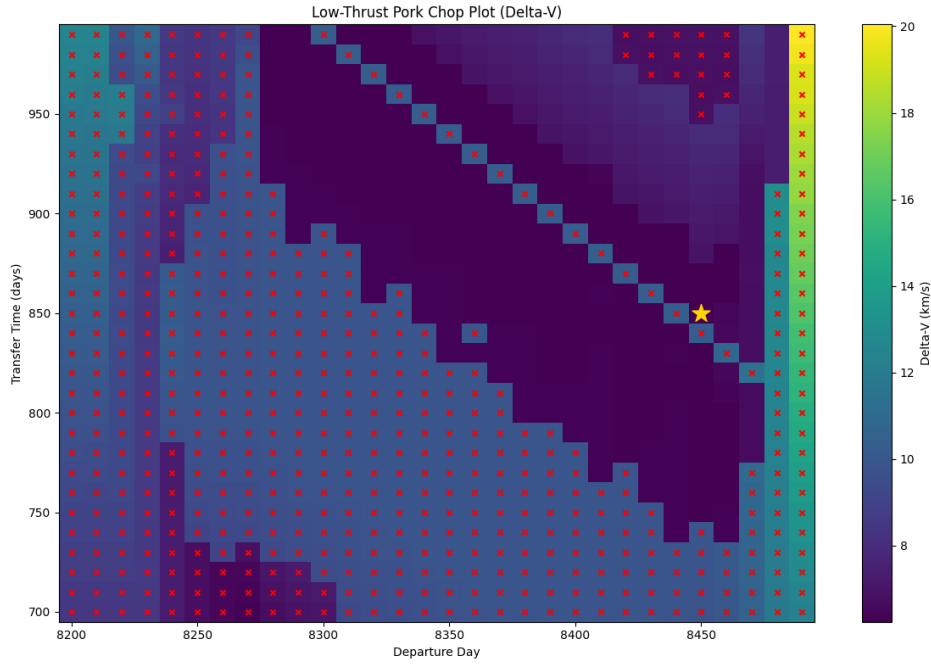


Figure 1: Porkchop plot for converged Q-law transfers in the time window of 2022-06-14 to 2023-04-10.

In Figure 1, the starred departure day and transfer time had the most favorable energy requirements. The 300-day range of departures was used as a search space for Q-law trajectory generation and launch date sensitivity analysis on further optimization.

Parameter Sweep Variables	
Maximum thrust	1.0-2.0 N
Specific impulse	1200-2400 N
Launch epochs	2022-06-14 to 2023-04-10

Table 2: Parameter sweep ranges for direct collocation

5 SEP Trajectory Optimization Results and Discussion

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5.1 Impulsive Solution

As a baseline for comparison, a classical two-impulse Hohmann transfer with a 1.85° plane change from Earth to Mars was computed assuming instantaneous velocity changes at perihelion and aphelion of the transfer ellipse. The spacecraft departs from a circular 400km LEO, executes the first Δv_1 maneuver to enter the transfer ellipse, coasts for approximately 259 days, and performs the second Δv_2 at Mars arrival to circularize into 300km MSO.

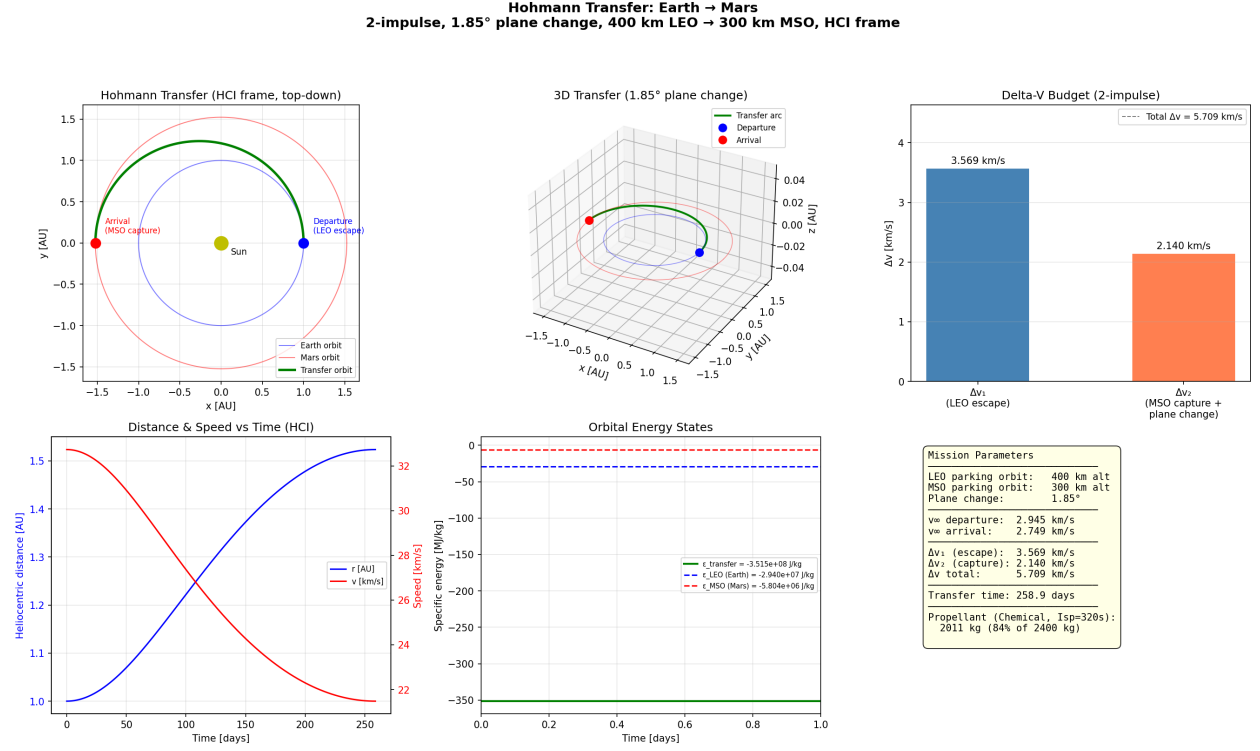


Figure 2: Hohmann transfer trajectory from Earth to Mars with 2-impulse maneuvers and 1.85° plane change. Top row shows the trajectory geometry in 2D (left), 3D (middle), and total propellant budget comparison (right). Bottom row displays distance versus time (left) and orbital energy evolution (right). Mission parameters are summarized in the table.

Figure 2 presents the Hohmann transfer solution with a 1.85° plane change executed at the second impulse. The total mission Δv is 5.709 km/s, split into 3.598 km/s for the departure burn and 2.140 km/s for the Mars orbit insertion maneuver. Using the Psyche spacecraft's specific impulse of $I_{sp} = 320$ s, the Tsiolkovsky rocket equation yields a propellant mass requirement of 2011 kg for the initial wet mass of 2400 kg, resulting in a final mass of approximately 389 kg after both maneuvers.

The trajectory exhibits the characteristic elliptical geometry of Hohmann transfers, with the spacecraft coasting unpowered for the majority of the 259-day transfer time. Orbital energy remains constant during the coast phase, as shown in the orbital energy state plot. The distance versus time profile displays the monotonic decrease from 1.0 AU at Earth departure to perihelion, followed by the climb to Mars at approximately 1.52 AU.

This impulsive baseline establishes the minimum time-of-flight benchmark of 259 days

for ballistic transfers but requires significantly more propellant than continuous-thrust SEP trajectories. The Hohmann solution serves as the reference case for evaluating the propellant savings achievable with low-thrust propulsion, traded against the increased mission duration inherent to spiral trajectories.

5.2 Low-Thrust Solution

First, the most fuel-optimal trajectory generated by Q-law in the launch window given in Section 4.3 is shown below in Figure 3. The control history is shown in Figure 4.

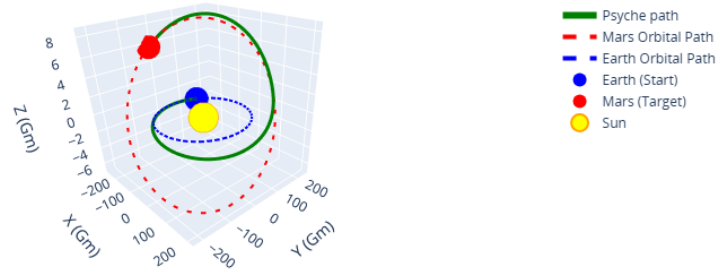


Figure 3: Q-law transfer departing 02-19-2023 with transfer time of 609 days.

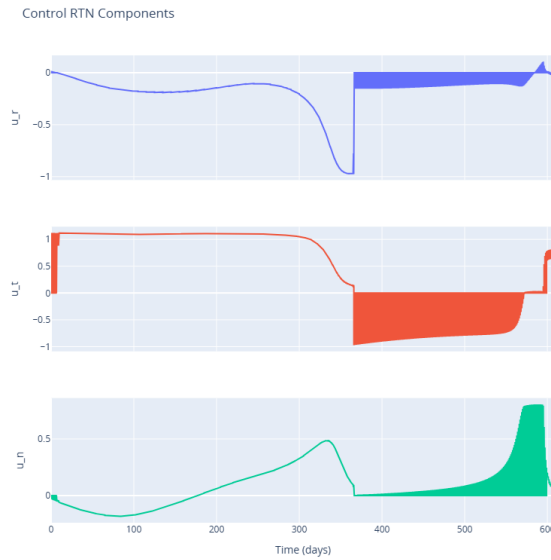


Figure 4: RTN thrust components for best Q-law transfer.

Hermite-Simpson collocation was run using the Q-law result as a seed trajectory. Using the objective function detailed in Section 3.1, a parametric sweep over weight combinations (W_1, W_2) was performed to characterize the tradeoff between propellant mass minimization and time-of-flight minimization. The mixed-weight cost function $J = W_1 \cdot m_{\text{prop}} + W_2 \cdot t_f$ allows continuous interpolation between pure mass-optimal and pure time-optimal solutions.

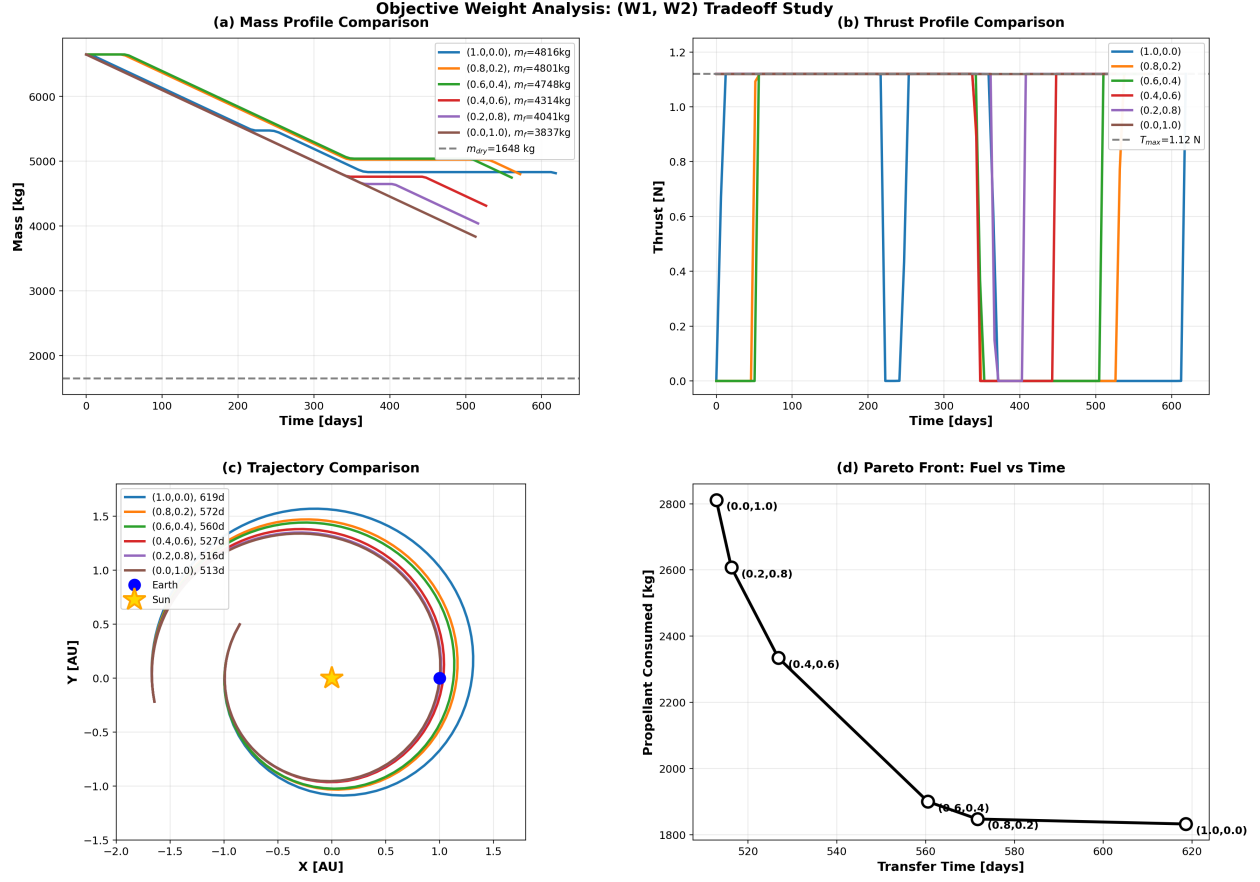


Figure 5: Objective weight analysis showing (a) final mass versus time-of-flight for varying (W_1, W_2) combinations, (b) thrust profile time histories, (c) trajectory geometries in the heliocentric frame, and (d) Pareto front representing non-dominated solutions trading propellant consumption against mission duration.

Table 3: Performance metrics across collocation objective weights. w_1 applies to fuel consumption and w_2 applies to transfer time.

(w_1, w_2)	Transfer time (d)	Δv (km/s)
(1.0,0.0)	619	0.662
(0.8,0.2)	572	0.717
(0.6,0.4)	560	0.913
(0.4,0.6)	527	2.606
(0.2,0.8)	516	3.760
(0.0,1.0)	513	4.675

Figures 5 and 3 present the results of this tradeoff study. In Figure 5, Panel (a) demonstrates that increasing the time-weight W_2 drives the optimizer toward faster transfers at the expense of higher propellant consumption. The baseline Q-law solution ($W_1 = 1.0, W_2 = 0$) prioritizes pure mass minimization, yielding approximately 609 days transfer time with min-

imal fuel consumption. As W_2 increases, transfer times decrease but propellant requirements rise. Panel (b) shows the corresponding thrust profiles: time-optimal solutions exhibit continuous full-thrust arcs, while mass-optimal solutions employ extended coasting periods to reduce propellant expenditure. The trajectory comparison in panel (c) reveals tighter spiral geometries for faster transfers. The Pareto front in panel (d) quantifies the non-dominated solution set, visually demonstrating the fundamental tradeoff between mission duration and propellant budget for the transfer problem. Figure 3 summarizes the transfer time and Δv accomplished by each objective for future comparison against other methods.

5.3 Sensitivity Analysis

5.3.1 Thrust Magnitude Variation

The effect of maximum available thrust was examined by scaling $T_{\max}$ from 1.0 N to 2.0 N while maintaining constant I_{sp} and dry mass.

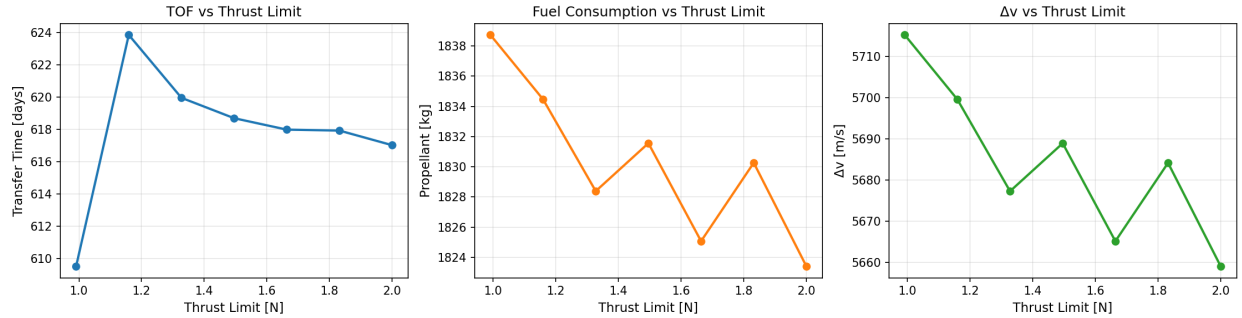


Figure 6: Thrust limit sensitivity showing transfer time, propellant consumption, and total Δv versus maximum thrust magnitude. Higher thrust enables faster transfers with minimal propellant penalty, demonstrating the benefit of increased thruster power.

The parametric sweep of thrust limit (1.0–2.0 N) reveals the tradeoff between mission duration and propellant efficiency (Fig. 6). Time of flight decreases monotonically from 624 days at 1.2 N to 617 days at 2.0 N, exhibiting diminishing returns above 1.4 N with only a 3-day reduction from 1.4 N to 2.0 N. The anomalously low TOF at 1.0 N (609 days) likely reflects optimizer convergence to a local minimum rather than the true optimal trajectory, suggesting numerical sensitivity in the low-thrust regime. Propellant consumption displays non-monotonic but generally decreasing behavior across the thrust range (1824–1839 kg), with the minimum occurring at 1.9–2.0 N. Similarly, total mission Δv decreases from 5715 m/s (1.0 N) to 5658 m/s (2.0 N), reflecting reduced gravity losses during spiral-out. The oscillatory patterns in propellant and Δv suggest sensitivity to Q-law guidance logic or multiple local optima, indicating that multi-start optimization strategies may be warranted.

The results delineate three operational regimes: a low-thrust regime (< 1.2 N) with potential convergence issues; a mid-thrust regime (1.2–1.6 N) offering balanced performance for practical mission design; and a high-thrust regime (> 1.6 N) with marginal improvements at increased power system mass cost. Excluding the noisy 1.0 N point, the data suggests

that 1.2–1.4 N represents an optimal design point, where adequate performance is achieved without overdesigning the propulsion system. For the Psyche-class baseline (1.0 N), upgrading to 1.5 N would reduce mission duration by approximately 6 days while consuming only 5 kg additional propellant.

5.3.2 Specific Impulse Sensitivity

The sensitivity of trajectory performance to specific impulse I_{sp} was investigated by varying I_{sp} from 1200 s to 2400 s while holding thrust power and other spacecraft parameters constant.

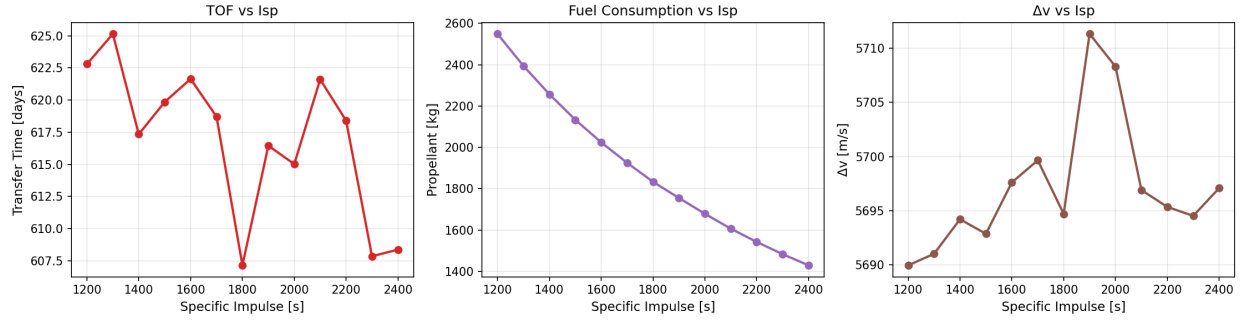


Figure 7: Specific impulse sensitivity study showing transfer time, propellant consumption, and total Δv as functions of I_{sp} . Higher I_{sp} monotonically reduces propellant mass but exhibits complex interactions with transfer time due to thrust-power coupling.

The parametric sweep of specific impulse (1200–2400 s) demonstrates a propellant-efficiency benefit for higher-Isp electric propulsion systems (Fig. 7). Propellant consumption decreases monotonically from 2570 kg at 1200 s to 1420 kg at 2400 s, representing a 45% reduction that directly reflects the Tsiolkovsky rocket equation’s linear relationship between Isp and propellant mass for a given Δv . In contrast, time-of-flight displays highly oscillatory behavior (607–625 days) with no clear trend across the Isp range, suggesting that the optimizer converges to different local trajectory solutions as Isp varies. Total mission Δv similarly exhibits non-monotonic variations (5690–5712 m/s, $\sim 0.4\%$ range), further indicating numerical sensitivity in the Q-law guidance or optimization convergence rather than fundamental mission physics.

The results highlight that while higher specific impulse provides unambiguous propellant savings, enabling a 1150 kg (45%) mass reduction when upgrading from 1200 s to 2400 s, it does not yield corresponding reductions in mission duration within the studied range. The lack of TOF correlation with Isp suggests that transfer time is dominated by orbital mechanics and thrust magnitude rather than exhaust velocity. The noisy convergence patterns underscore the need for robust multi-start optimization strategies when exploring the Isp design space. For the Psyche baseline ($I_{sp} \sim 2000$ s), these results validate the selection of Hall-effect thruster technology, which achieves near-optimal propellant efficiency within the practical specific impulse range for xenon propellant systems.

5.3.3 Launch Window Analysis

Launch date sensitivity was evaluated across a 120-day window centered on the nominal departure epoch (J2000 day 8450, corresponding to February 19, 2023).

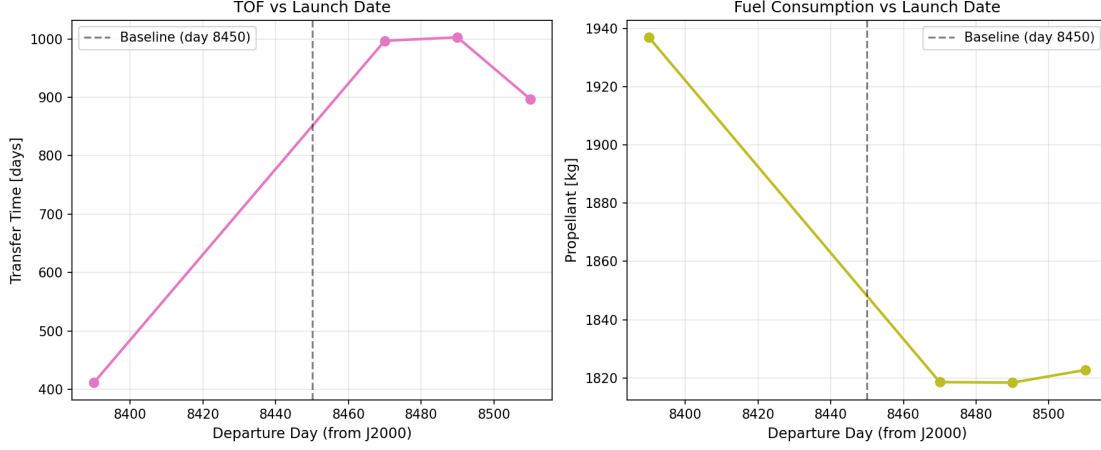


Figure 8: Launch window sensitivity showing transfer time and propellant consumption versus departure date. The baseline solution (dashed vertical line) lies near the optimal launch opportunity, demonstrating strong dependence on Earth-Mars relative phasing.

The parametric sweep of departure date (J2000 days 8390–8510) reveals strong sensitivity to Earth-Mars phasing, though sparse data reflects significant optimizer convergence challenges across this window (Fig. 8). Only four successful trajectories were obtained across the 120-day span, with a notable 60-day gap between the first converged solution (day 8390) and the baseline (day 8450), likely indicating that intermediate launch dates encounter unfavorable orbital geometry incompatible with the Q-law guidance structure or optimizer initial guess. The converged solutions show dramatic variation in time of flight (410–1000 days) and more modest propellant differences (1818–1937 kg, 6% range), though the limited sampling precludes confident interpolation of trends between points.

The sparsely sampled results suggest a fundamental challenge in low-thrust trajectory optimization across launch windows: certain departure phasings may preclude convergence to feasible solutions within the Q-law feedback control framework, particularly when Earth-Mars geometry requires trajectory geometries far from the initial guess structure. The early departure (day 8390) achieves a 410-day transfer with 1937 kg propellant, while near-baseline departures (days 8450–8490) require 900–1000 days but only 1818–1820 kg propellant—a potential 400–500 day reduction at 100 kg cost. However, the convergence gaps indicate that a more robust optimization approach would be necessary to fully characterize the launch window tradeoffs and identify whether intermediate departure dates offer viable compromise solutions.

5.4 Interpretation

Table 4: Performance metrics across five generated trajectories.

Trajectory	Transfer time (d)	Δv (km/s)
Hohmann	259	0.507
Q-law seed	609	6.24
Fuel-optimized	619	0.662
Time-optimized	513	4.537

SEP offers clear mass savings but at the cost of longer mission durations. The analysis evaluated the primary operational tradeoff between significant reductions in propellant mass and overall increases in transfer time, revealing that an optimized low-thrust transfer can use as little as 12% of the Δv required for an impulsive Hohmann transfer. However, this propellant efficiency comes at the expense of mission duration: the best low-thrust transfer time reaches 1.98 times the impulsive transfer time of 259 days. By demonstrating these operational benefits using a low-thrust SEP system modeled on Psyche’s xenon gas parameters for Earth-Mars transfers, the study confirmed that the quantified propellant savings can justify the extended transfer duration under a heliocentric 2-body model. The results also show that trajectories become more fuel-efficient with higher specific impulse, supporting the prioritization of efficiency over power in low-thrust propulsion systems. However, it is important to contextualize these findings within the project’s limitations. The assumption of continuous solar power ignores eclipsing effects, and the exclusion of planetary gravity assists may artificially simplify the efficiency of the optimized trajectory, potentially overstating the performance gains achievable in a real mission scenario.

6 Aerobraking Trajectory Modeling

Author(s): Annika Vaidyanathan

6.1 Introduction

The first part of this project modeled the SEP transfer and ending with a spacecraft in an initial elliptical orbit around Mars. Although this means the spacecraft is captured by Mars gravity, it is still in a noncircular orbit that is too high for Martian science missions than a target circular orbit. Therefore, this section focuses on modeling how a spacecraft can aerobrake to a final circularized orbit, rather than using a purely propulsive burn, with the goal of reducing propellant requirements. Following the mission architecture used in both MAVEN and Mars Global Surveyor (MGS), the goal of this section of the paper is to model each atmospheric pass and evaluate how drag removes orbital energy, lowers the orbit, and moves the spacecraft closer to its final circularized orbit. Here we also consider the limits that constrain aerobraking performance, including atmospheric density, spacecraft mass, drag area, peak dynamic pressure, and thermal loading.

6.2 Martian Orbit Path Selection

For all aerobraking simulations, we selected initial and final orbits based on the Mars Global Surveyor mission. The initial orbit is an elliptical post-insertion capture orbit, with a periapsis altitude of 262.9 km, an apoapsis altitude of 54,025.9 km, an inclination of approximately 93°, and an orbital period of about 45 hours [1]. The target final orbit is a nearly circular low Mars orbit at an altitude of 378 km with an inclination of 90°, which is representative of a typical science orbit [1]. Although the starting orbit does not necessarily align with the final orbit of the SEP sections, these starting and final orbits were selected to parallelize the progress of this paper, and are still representative of an overall trajectory that could be used to enter the Martian atmosphere.

6.3 Modeling of Atmospheric Passes

The overall goal of this section is to develop the aerobraking, trajectory model in Python, and calculate the metrics for trajectory selection (fuel consumption, transfer time, number of passes).

The code starts by calculating orbital geometry, using the periapsis (r_p) and apoapsis (r_a) to calculate the semimajor axis (a) and eccentricity (e) using:

$$a = \frac{r_p + r_a}{2}, \quad e = \frac{r_a - r_p}{r_a + r_p}, \quad (27)$$

The velocity of the spacecraft is then calculated using the vis-viva equation:

$$v = \sqrt{\mu \left(\frac{2}{r} - \frac{1}{a} \right)}, \quad (28)$$

where μ is the Martian gravitational parameter. For the final circular orbit, the circular velocity is

$$v_c = \sqrt{\frac{\mu}{r}}. \quad (29)$$

The direct propulsive transfer, which is the baseline/traditional case, is calculated as an impulsive maneuver at apoapsis, followed by a final circularization burn. When an inclination change is included, the impulsive burn magnitude is computed using:

$$\Delta V = \sqrt{v_1^2 + v_2^2 - 2v_1v_2 \cos(\Delta i)}, \quad (30)$$

where v_1 and v_2 are the pre-burn and post-burn orbital speeds and Δi is the inclination change. The final circularization burn is calculated as

$$\Delta V_{circ} = |v_c - v_{orbit}|, \quad (31)$$

where v_{orbit} is the spacecraft speed at the circularization point and v_c is the required circular velocity. The aerobraking code however, starts with an impulsive entry burn that lowers periapsis from the initial capture orbit into the upper Martian atmosphere. After this burn, the spacecraft repeatedly passes through the atmosphere near periapsis. During

each pass, atmospheric drag removes orbital energy, which causes apoapsis to decrease. The code uses a pass-by-pass energy propagation method where each orbit is treated as Keplerian while not near periapsis, and the energy loss due to atmospheric drag is integrated over a specified range of true anomaly values (which were optimized through iterations of results), centered around periapsis. This simplification was made because most of the aerobraking happens around periapsis, when the spacecraft is closest to the atmosphere.

For each atmospheric pass, the code evaluates the spacecraft trajectory over a true anomaly range, calculating the orbital radius using:

$$r(f) = \frac{a(1 - e^2)}{1 + e \cos f} \quad (32)$$

and the altitude. Aerodynamic drag is modeled using the spacecraft velocity relative to the rotating atmosphere, since aerodynamic deceleration depends on the spacecraft velocity relative to the spinning velocity of the atmosphere. The drag acceleration is calculated as:

$$\mathbf{a}_{drag} = -\frac{1}{2}\rho \frac{C_D A}{m} \|\mathbf{v}_{rel}\| \mathbf{v}_{rel}, \quad (33)$$

where C_D is the drag coefficient, A is the effective drag area, m is spacecraft mass, and $\mathbf{v}$ is the spacecraft velocity. For all spacecrafts, the drag coefficient was assumed to be 2.

The total energy lost over one atmospheric pass is found by numerically integrating the energy loss rate. Because the code discretizes the trajectory using true anomaly rather than time, this integral is represented as:

$$\Delta E = \int \frac{dE}{dt} \frac{dt}{df} df \quad (34)$$

The conversion between true anomaly and time comes from angular momentum,

$$h_{ang} = r^2 \frac{df}{dt} \quad (35)$$

Therefore, the energy loss per pass becomes

$$\Delta E = \int \left(-\frac{1}{2}\rho \frac{C_D A}{m} v^3 \right) \frac{r^2}{h_{ang}} df \quad (36)$$

Which was solved using the trapezoidal rule (for simplicity) over the true anomaly range. After each pass, the orbit is updated using:

$$\epsilon = -\frac{\mu_M}{2a} \quad (37)$$

The new apoapsis is calculated and becomes the starting condition for the next atmospheric pass. Once the apoapsis is at the target orbit altitude, a final impulsive burn is applied to fully circularize the orbit. Note that the entry and circularization burns are treated as impulsive.

6.4 Modeling J_2

Instead of modeling the spacecraft acceleration as purely two-body gravity plus atmospheric drag, the model also includes the effect of Mars' oblateness by including J_2 . J_2 acceleration is computed from the spacecraft's position relative to Mars and added to the gravitational acceleration (from the two-body problem) before the drag acceleration is considered. This allows the model to account for small deviations from an idealized gravity field, caused by Mars' equatorial bulge, implemented using:

$$\vec{a}_{J_2} = \frac{3J_2\mu_M R_M^2}{2r^5} \begin{bmatrix} x \left(5\frac{z^2}{r^2} - 1 \right) \\ y \left(5\frac{z^2}{r^2} - 1 \right) \\ z \left(5\frac{z^2}{r^2} - 3 \right) \end{bmatrix}, \quad (38)$$

where μ_M is the gravitational parameter of Mars, R_M is the mean radius of Mars, r is the distance between the spacecraft and the center of Mars, and x , y , and z are components the position of the spacecraft.

6.5 Atmospheric Density Model

The aerobraking model uses a simplified Martian atmospheric density model that was manually derived from MGS flight data. Since atmospheric drag is directly proportional to density, the accuracy of the density model strongly affects the predicted energy loss, as well as any calculated dynamic pressure and heating rate. The number density measured by MGS during its aerobraking campaign was converted to mass density under the assumption that the Martian atmosphere is 100% CO_2 .

The MGS mission occurred during a period of increasing solar activity in solar cycle 23, so the atmospheric profiles include both lower and higher solar activity cases [12]. This allows the model to represent a range of possible atmospheric densities rather than just relying on a single fixed profile.

To fill any gaps in the data, the MGS density measurements were fit using the traditional exponential atmosphere equation:

$$\rho(h) = \rho_{\text{ref}} e^{-(h-h_{\text{ref}})/H}, \quad (39)$$

where $\rho(h)$ is the atmospheric density at altitude h , ρ_{ref} is the reference density at altitude h_{ref} , and H is the atmospheric scale height.

For each altitude, the model evaluates the fitted density profiles and uses the median value as the representative atmospheric density. The shaded profile range shown in the density plot represents the spread between the minimum and maximum fitted profiles. Seen in Figure ??, the atmospheric model neglects atmospheric density above 200 km, with the light blue shaded region representing the altitudes we are aerobraking in. Seen in the uncertainty of the dark blue region, this atmospheric model is good at capturing general trends, but does not capture full variations in latitude, longitude, local solar time, dust conditions, or seasonal atmospheric changes.

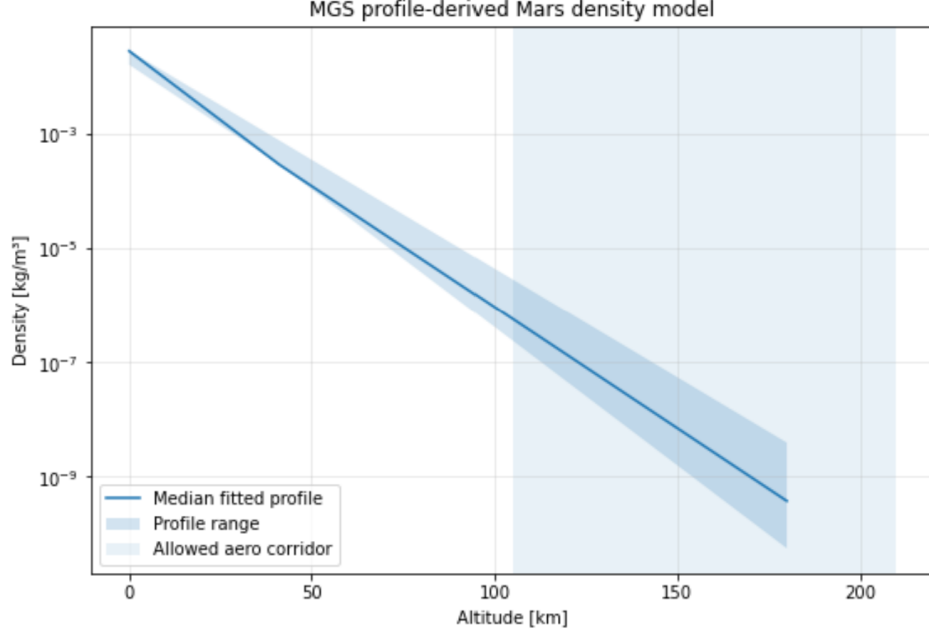


Figure 9: The atmospheric density model of Mars, derived from MGS flight data

6.6 Dynamic Pressure and Thermal Constraints

Finally, the aerobraking model was made to conciser the dynamic pressure and thermal constraints of these different spacecraft. Since the spacecraft experiences the largest atmospheric loading near periapsis, where both velocity and atmospheric density are highest, the constraint checks are applied primarily around the periapsis region. The first constraint considered is the peak dynamic pressure, which is calculated as:

$$q = \frac{1}{2} \rho v_{rel}^2, \quad (40)$$

where q is the dynamic pressure, ρ is the local atmospheric density, and v_{rel} is the spacecraft velocity relative to the atmosphere.

The code also adjusts the effective drag area based on the dynamic pressure encountered during the pass. At lower dynamic pressures, the spacecraft uses a larger drag area to increase atmospheric braking. At higher dynamic pressures, the model reduces the spacecraft's effective drag area to limit aerodynamic loading. This approximates controlled attitude-based drag changes while aerobraking, where the spacecraft orientation can be adjusted to expose more or less area to the flow. In implementation, a spacecraft had a defined minimum and maximum drag area, with linear curve to fit the areas in between.

To address the thermal loading constraints during aerobraking, the model estimates the stagnation-point convective heating rate using a Sutton-Graves equation:

$$\dot{q} = k \sqrt{\frac{\rho}{R_n}} v_{rel}^3, \quad (41)$$

where $\dot{q}$ is the convective heat rate, k is an empirical heating coefficient, ρ is atmospheric density, R_n is the effective nose radius, and v_{rel} is the atmospheric-relative velocity. This

equation shows that heating is especially sensitive to velocity because it scales with v_{rel}^3 . As a result, even small changes in the atmospheric pass conditions can significantly affect the thermal environment. For each pass, the code selects a periapsis altitude that satisfies both the dynamic pressure and heating constraints. The model chooses the lowest safe periapsis altitude by aiming for 85% of the dynamic pressure and heating constraints.

7 Aerobraking Parametric Mass Study Results

Author(s): Annika Vaidyanathan

7.1 Aerobraking Parametric Mass Study

Using the aerobraking simulations, the next goal of this project is to assess how spacecraft design choices affect aerobraking performance at Mars through a parametric mass study. This study compares how different vehicle configurations behave in the aerobraking trajectory model, comparing the number of atmospheric passes, total time of flight, peak dynamic pressure, and ΔV savings relative to a fully propulsive trajectory. By varying the spacecraft configuration, the study aims to help identify which characteristics make aerobraking more efficient and which designs require longer, more gradual campaigns.

For the study, we choose three Mars orbiter configurations: an MGS-like orbiter, an Odyssey-like orbiter, and a MAVEN-like orbiter. These spacecraft were chosen because they represent different masses, drag areas, and aerobraking-relevant geometries from actual Mars missions. The MGS-like case provides a direct comparison to the Mars Global Surveyor mission architecture used for the starting and final orbit assumptions. The Odyssey-like case represents a lighter spacecraft with a smaller effective drag area. The MAVEN-like case represents a larger-area, deep-dip configuration, making it useful for studying how increased drag area affects aerobraking trajectories. These three configurations provide a realistic range of spacecraft properties which are outlined in the table below.

Table 5: Spacecraft parameters used in the aerobraking parametric study

Spacecraft	Mass	Drag Area Range	Source
MGS-like orbiter	1060 kg	$A_{\min} = 4 \text{ m}^2$	[10]
		$A_{\text{nom}} = 10 \text{ m}^2$	
		$A_{\max} = 18 \text{ m}^2$	
Odyssey-like orbiter	730 kg	$A_{\min} = 3.5 \text{ m}^2$	[11]
		$A_{\text{nom}} = 8 \text{ m}^2$	
		$A_{\max} = 14 \text{ m}^2$	
MAVEN-like orbiter	809 kg	$A_{\min} = 8 \text{ m}^2$	[12]
		$A_{\text{nom}} = 18 \text{ m}^2$	
		$A_{\max} = 26.54 \text{ m}^2$	

It is also important to note that during flight, MGS and Odyssey were primarily constrained by dynamic pressure, so representative pressure limits were set to 0.6 N/m^2 and 0.3 N/m^2 respectively [13]. MAVEN had a higher dynamic pressure constraint, set to

0.8 N/m² in the code. Additionally, all three spacecrafts were constrained by a maximum heating rate of 2500 W/m² based on previous flight data [14].

7.2 Results and Discussion

The table below summarizes the bulk of the final results for each of the three configurations. It is important to note that all three trajectories start and end using the previously defined starting and ending orbits from the MGS mission.

Table 6: Aerobraking performance comparison for the three spacecraft configurations

	MGS	Odyssey	MAVEN
Passes	612	532	212
Time of Flight	238.9 days	207.7 days	83.0 days
$\Delta V_{\text{aerobraking}}$	0.082 km/s	0.081 km/s	0.077 km/s
$\Delta V_{\text{traditional}}$	1.26 km/s	1.26 km/s	1.25 km/s
Peak dynamic pressure	0.383 Pa	0.383 Pa	0.510 Pa

Seen in the table, results show that all three spacecraft configurations achieve significant propulsive ΔV savings relative to the traditional fully propulsive baseline. Overall, MGS and the Odyssey like spacecrafts have similar metrics, with MAVEN having a shorter duration and requiring less ΔV .

Figure 10 shows the periapsis altitude as a function of the time of flight. The periapsis altitude is adjusted throughout the aerobraking campaign in order to keep each pass within the dynamic pressure and thermal constraints. Overall, figure 10 shows how the MGS-like and Odyssey-like configurations have similar trajectories, likely due to their similar mass, drag ratios, and also their dynamic pressure constraints. The MAVEN trajectory seen in Figure 10 is a much more aggressive aerobraking profile, likely because of its larger drag area and higher allowable dynamic pressure. Interestingly, the change in the trajectory trend at around 75 days for both the Odyssey-like and MGS-like configurations is a product of the dynamic pressure constraints, aligning with NASA’s data that their aerobraking trajectories were constrained by dynamic pressure.

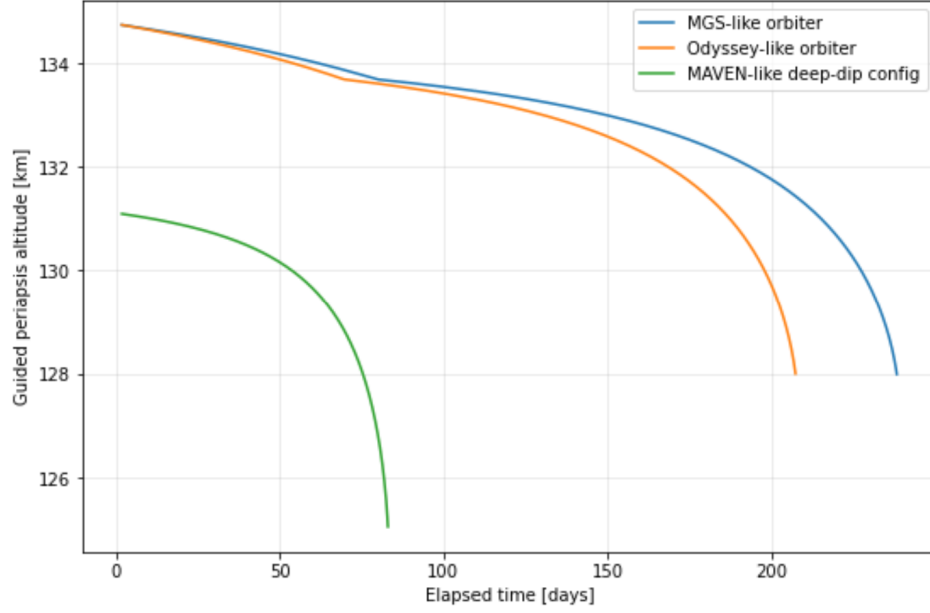


Figure 10: Trajectory Shape Visualization for all three configurations

Figure 11 compares the aerobraking trajectory time of flight and spacecraft mass. The MGS-like orbiter has the most mass and also has the longest campaign duration, consistent with the expectation that a heavier spacecraft is harder to slow down through drag. However, the MAVEN-like case is heavier than the Odyssey-like case but still reaches the target orbit faster. This trend likely is because MAVEN has a significantly larger drag area, which lowers its effective ballistic coefficient more than its higher mass impacts the trajectory. Additionally, MAVEN has much more extreme maximum and minimum drag areas. This means that it is able to better optimize its drag area used to be as close to the dynamic pressure constraint as possible without going over it. Therefore, the results show that aerobraking performance depends on the combined effect of mass, drag coefficient, and reference area, rather than mass alone. Based on this, future missions should look at optimizing a mass to drag area, and configurations that give a wider range of drag areas to be used for aerobraking.

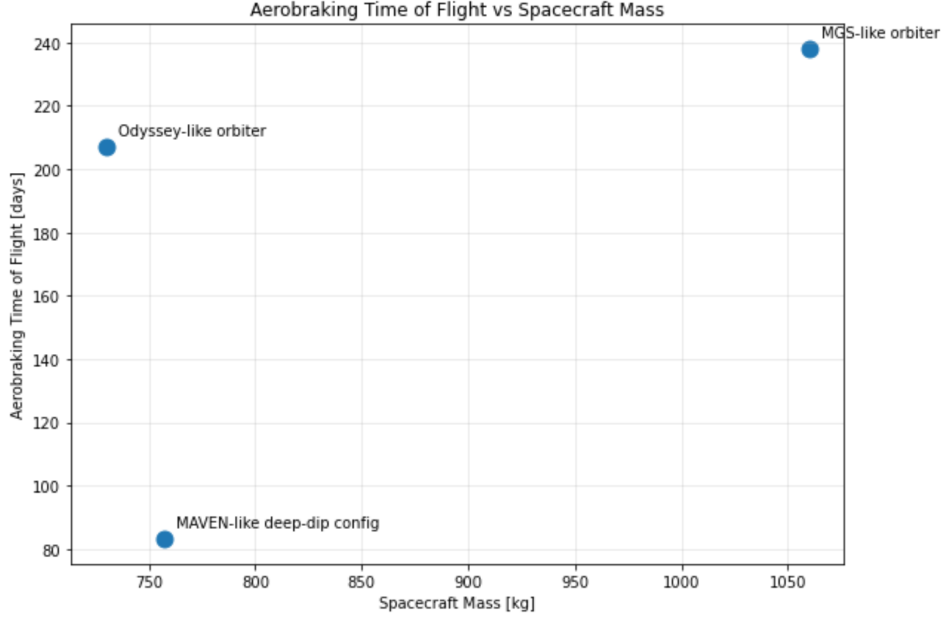


Figure 11: Time vs. Mass

Figure 12 highlights the tradeoff between ΔV savings and time of flight, showing that there is no direct correlation between ΔV savings and time of flight. The aerobraking trajectories for each configuration save approximately similar amounts of ΔV compared to the traditional propulsive transfer. However, the time required to achieve these savings varies substantially between configurations. The MAVEN-like deep-dip configuration reaches the target orbit in the shortest amount of time, while the MGS-like and Odyssey-like configurations require longer aerobraking campaigns. This result is primarily caused by differences in spacecraft drag area and ballistic coefficient. The MAVEN-like configuration has a much larger effective drag area, allowing it to remove more orbital energy during each atmospheric pass. As a result, it requires fewer passes and less total time to lower apoapsis to the target orbit.

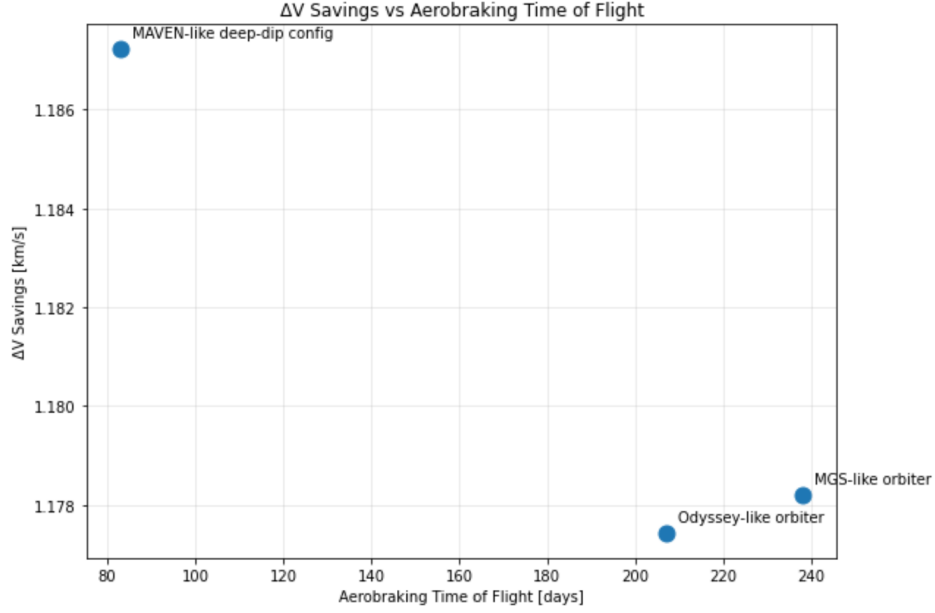


Figure 12: Delta-V vs. Time of Flight

Finally, combining the approach of the SEP section of this paper and the aerobraking section, Figure 13 presents the ΔV requirements for a traditional propulsive trajectory, aerobraking trajectory, and the low thrust (SEP) trajectory for each of these three configurations. Note that all three of these trajectory methods are transfer trajectories from the starting and ending MGS orbits. From Figure 13 it is clear that both aerobraking and the low thrust SEP trajectories save a significant amount of ΔV . While the aerobraking trajectory does assume chemical propulsion and the low thrust system assumes electrical propulsion, a combination of these two approaches would provide an extremely ΔV cost effective solution. Most spacecrafts that have electric propulsion, also have chemical propulsion capabilities, so the results of this study illustrate that combining these two approaches is feasible and optimal in reducing ΔV requirements to Mars.

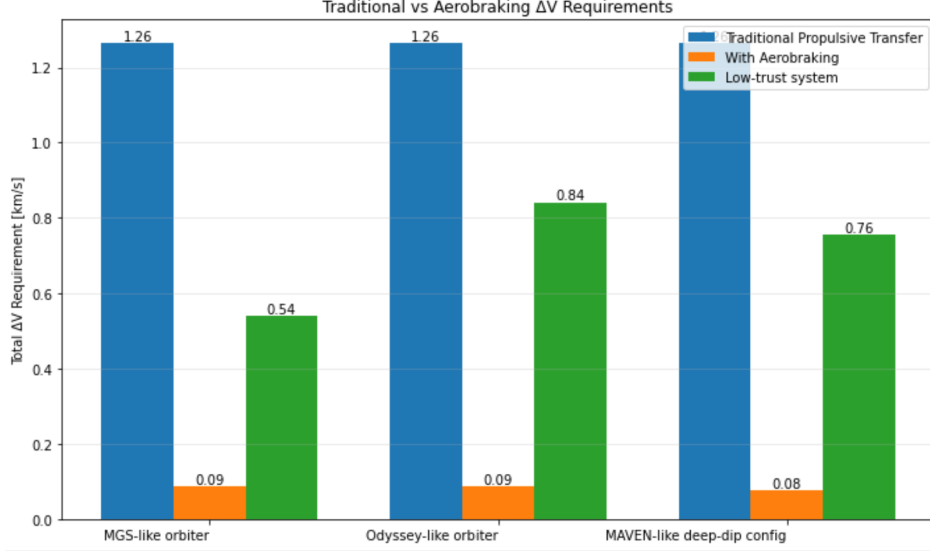


Figure 13: ΔV requirements for a traditional propulsive trajectory, aerobraking trajectory, and the low thrust (SEP) trajectory

8 Conclusions and Future Work

Author(s): Annika Vaidyanathan, Megan Tseng

8.1 General Key Findings

The low-thrust solution is shown to significantly reduce the energy requirements for an Earth-Mars interplanetary transfer. Optimal control was necessary for solving this problem because it provided continuous steering to the propulsion system and ensured maximum fuel efficiency. First, Q-law control provided an initial guess trajectory that chose the "greedy" optimal thrusting vector at every integration step. While the Q-law result did not improve upon the Hohmann fuel and time outcomes, collocation was able to discretize the trajectory and achieve a much more fuel-optimal result. By thrusting at low accelerations over long durations, low-thrust maneuvers are able to reduce total power consumption and reduce the necessary fuel carried on the spacecraft. It also solidifies the benefit of combining direct and indirect optimization methods to reach greater optimality.

Using Psyche-like parameters to build the spacecraft in this simulation also provided a proof-of-concept for a mission that utilized a Mars flyby. Theoretically, interplanetary missions of this duration can reasonably consider a similar low-thrust trajectory using existing hardware like Psyche's SEP system.

The aerobraking model successfully modeled the orbital energy dissipation (energy loss per pass), considering gravity, J2, dynamic pressure, heat rate, and atmospheric drag, achieving the goal of developing a simplified comparative model for aerobraking trajectories. From the results, Mars' central gravity governs the main orbital motion, while the J2 accounts for long-term precession and drift in the orbital elements caused by Mars' non-spherical gravity

field. Atmospheric drag is also the key non-conservative force in the model, gradually shrinking the orbit by lowering apoapsis over repeated passes. Overall, the results demonstrate that aerobraking is highly effective for reducing propulsive ΔV after Mars capture. However, this benefit comes with operational tradeoffs. Aerobraking requires many repeated atmospheric passes, careful periapsis targeting, and management of dynamic pressure and thermal constraints. Among the three configurations studied, the MAVEN-like deep-dip configuration performed the best according to the Python model, likely because with our considerations, its larger drag area allows it to dissipate orbital energy more quickly. The MGS-like and Odyssey-like cases still achieve comparable ΔV savings, but they take more time and passes because each pass removes less energy. These results suggest that spacecraft geometry and ballistic coefficient are critical design parameters for aerobraking mission performance.

Together, the results of the SEP and Aerobraking studies show that both methods have potential for reducing ΔV requirements for missions to Mars. We therefore, propose missions look into a combined electric and chemical propulsion system that can levy both these methods of propulsion to increase payload capacity to Mars!

8.2 SEP Future Work

Since the low-thrust trajectory optimization is highly sensitive to the initial seed, future work naturally includes developing a more robust Q-law control scheme. While direct collocation was effective at extending the interplanetary Q-law trajectory to Earth escape and Mars entry, creating controllers for all three stages of the transfer and creating a full patched-conics trajectory could yield improved collocation results. In addition, rigorous optimization of tunable Q-law parameters like throttle values and orbital element tolerances could provide launch windows with better convergence. Instead of using one numerical coast threshold, more efficient thrust-coast arcs could be achieved by analytically determining thrust effectiveness thresholds at each integration step.

Orbit propagation in the low-thrust simulation was also simplified, with two-body dynamics and no perturbations besides applied thrust. Usage of three- or n-body dynamics and inclusion of perturbations like J2 would improve simulation fidelity for comparison against an existing mission like Psyche. Inclusion of atmospheric drag in the Earth escape and Mars entry stages would provide more realistic Δv estimates and better comparison against the aerobraking solution for both the impulsive and low-thrust solutions.

Furthermore, choosing other planets in our solar system to test this low-thrust control scheme could help to analyze the benefits of each trajectory in farther or closer bodies. Especially since available power for SEP decreases with distance from the Sun, it would be interesting to see when the benefits of low-thrust SEP trade off with the penalty of travelling too far away from the Sun.

8.3 Aerobraking Future Work

Future work for this project should focus on increasing the fidelity and validation of the aerobraking model. First, the Python model should be validated and compared to aerobraking trajectory modeling in STK (Astrogator). The drag model could also be improved by including more detailed spacecraft geometry and aerodynamic behavior, rather than relying

only on simplified drag areas and constant drag coefficients. Future versions of the model should use MarsGRAM or a similar atmospheric tool to better represent the variability of the Martian atmosphere. Density uncertainty and Monte Carlo simulations should also be added to test how sensitive the aerobraking trajectory is to atmospheric variability, especially because small changes in density can strongly affect drag, heating, dynamic pressure, and total flight time.

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A Heliocentric Modified Equinoctial Elements

Using Cartesian state vectors $\{\mathbf{r}, \mathbf{v}\}$, Heliocentric MEE coordinates $\{\mathbf{p}, \mathbf{f}, \mathbf{g}, \mathbf{h}, \mathbf{k}, \mathbf{L}\}$ are obtained by:

1. Semi-latus rectum $\mathbf{p}$

$$p = \frac{||\mathbf{h}||^2}{\mu} \quad (42)$$

2. Eccentricity parameters $\mathbf{f}, \mathbf{g}$

$$f = e_x, \quad g = e_y \quad (43)$$

3. Inclination parameters $\mathbf{h}, \mathbf{k}$

$$h = \frac{h_x}{||\mathbf{h}|| + h_z}, \quad k = \frac{-h_y}{||\mathbf{h}|| + h_z} \quad (44)$$

4. True longitude $\mathbf{L}$

$$\alpha = 1 + h^2 + k^2 \quad (45)$$

$$L = \text{atan2} \left(\frac{r_y(1 + h^2 - k^2) - r_x(2hk) + r_z(2h)}{\alpha}, \frac{r_x(1 - h^2 + k^2) + r_y(2hk) + r_z(2k)}{\alpha} \right) \quad (46)$$

where $\mathbf{h}$ is the angular momentum vector and $\mathbf{e}$ is the eccentricity vector.

B Q-law parameters

Table 7: Tunable Q-law parameters for throttling, phasing, and arrival.

Parameter	Value
Coast threshold	0.01
Nominal throttle	0.8
Phasing tolerances	3%
Arrival tolerances	0.5%